



12. Packed bed reactors are commonly used with:
 A) Liquid substrates only B) Air as medium
 C) Immobilized enzymes or cells D) Free enzymes in solution
13. The unit of power consumption in an agitation system is:
 A) kW or W B) kg/m³
 C) m³/min D) Pa
14. The impeller most commonly used in aerobic fermenters is:
 A) Helical ribbon B) Pitched blade turbine
 C) Paddle stirrer D) Rushton turbine
15. If a fermenter has a volume of 100 L and receives 200 L/min of air, the aeration rate is:
 A) 2 vvm B) 0.5 vvm
 C) 1.5 vvm D) 0.2 vvm
16. Aeration rate must be adjusted based on
 A) Agitator Speed B) Rheological conditions (e.g., viscosity)
 C) pH D) Tank shape only
17. RTDs work on the principle that:
 A) Resistance increases with temperature B) Voltage decreases with pressure
 C) Current remains constant D) Capacitance varies with temperature
18. Calibration of DO probe is done using:
 A) Alcohol B) Dry gas only
 C) Distilled water D) 0% and 100% oxygen saturation
19. The transfer function is a mathematical representation of a system in the:
 A) Time domain B) Laplace domain
 C) Frequency domain D) Real number domain
20. In a first-order system, the time constant (τ) determines:
 A) Final value of output B) Gain of the system
 C) Time to reach 63.2% of the final value D) Input resistance

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the major components of a bioreactor and explain the specific purpose each component serves in ensuring optimal fermentation conditions.	8	L*2	1	3
	b) Explain the steps involved in validation of cleaning for industrial fermenter.	8	L2	2	3
2.	a) Explain the factors to consider when selecting materials of construction for a bioreactor. How do corrosion and the need for protective coatings or linings influence material choice?	8	L2	1	3
	b) Explain the step-by-step operation of a batch fermenter, starting from initial cleaning to final product harvesting. Why is each step crucial for successful fermentation?	8	L2	2	3
3.	a) List and explain different types of impellers used in fermenter. Also write the selection criterion for impellers in bioprocess industries.	8	L2	1	3
	b) List and explain the design guidelines to be followed to aid the cleaning of bioreactor vessel and pipeline network	8	L2	2	3
Unit – II					
4.	a) Derive an equation to design an ideal Chemostat for microbial growth. How is the dilution rate defined, and what role does it play in steady-state operation?	8	L3	3	3
	b) Define the term 'volumetric aeration rate (vvm)'. How is vvm calculated, and how do rheological properties of the fermentation medium affect vvm?	8	L3	4	2

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| 5. a) | Describe the principle and operation of a recycle reactor. Using suitable expression, explain how recycling improves substrate conversion and reactor performance? | 8 | L3 | 3 | 3 |
| b) | Compare different techniques for measuring k_{La} in bioreactors. Which method is most used and why? | 8 | L3 | 4 | 2 |
| 6. a) | A strain of <i>Escherichia coli</i> has been genetically engineered to produce human protein. A batch culture is started by inoculating 12g cells into a 100L bubble-column fermenter containing 10g/L glucose. The maximum specific growth rate of the culture is 0.9/h; the biomass yield from glucose is 0.575g/g. | | | | |
| (i) | Estimate the time required to reach stationary phase. | | | | |
| (ii) | What will be the final cell density if the fermentation is stopped after only 70% of the substrate is consumed? | 8 | L3 | 3 | 3 |
| b) | A bioprocess in which oxygen requirement for aerobic cell culture is maintained by continuous sparging of sterile air. Derive an expression to determine the rate of mass transfer between liquid culture and gaseous oxygen. | 8 | L3 | 4 | 2 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Discuss the operational principle of a glass electrode used for pH measurement in bioprocesses. What factors affect its accuracy, and how can these be mitigated through maintenance practices? | 8 | L2 | 5 | 3 |
| b) | Differentiate between P, PI, PD, and PID controllers in terms of their functionality and applications. Provide examples where each controller type would be most effective in a fermentation process. | 8 | L2 | 5 | 3 |
| 8. a) | Describe how turbidity measurements can be used to estimate biomass in a bioreactor. What are the limitations of this method, and how can sensor calibration improve its reliability? | 8 | L2 | 5 | 3 |
| b) | Define and derive the transfer function for a first-order system in the Laplace domain. Explain how such a transfer function can be used to analyze the transient behavior of a sensor responding to a step input. | 8 | L3 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
